



Contextra: A Study-Structure-Aware Multimodal Agent Framework for Sample-Level Materials Knowledge Extraction

Seyong Im, Jisu Kim, and Seungwu Han*

Department of Materials Science and Engineering, Seoul National University, Seoul, 08826, Korea

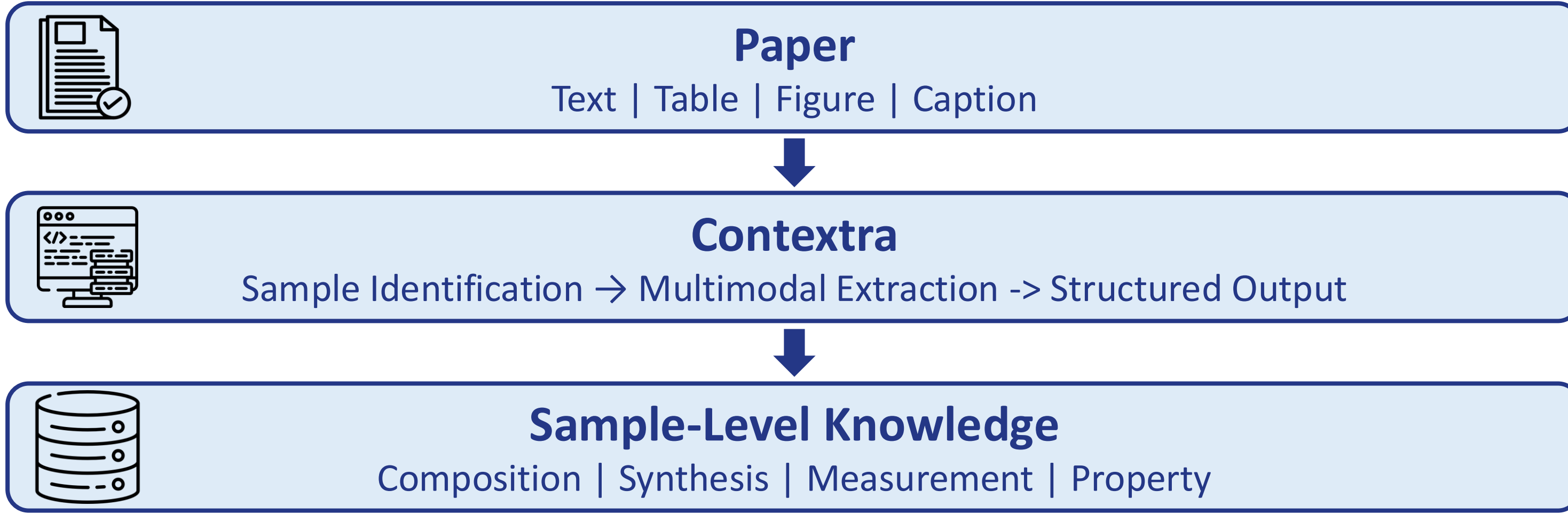
Introduction

From Scattered Literature to Sample-Level Materials Knowledge

Materials knowledge exists in papers, but not in a research-ready structure

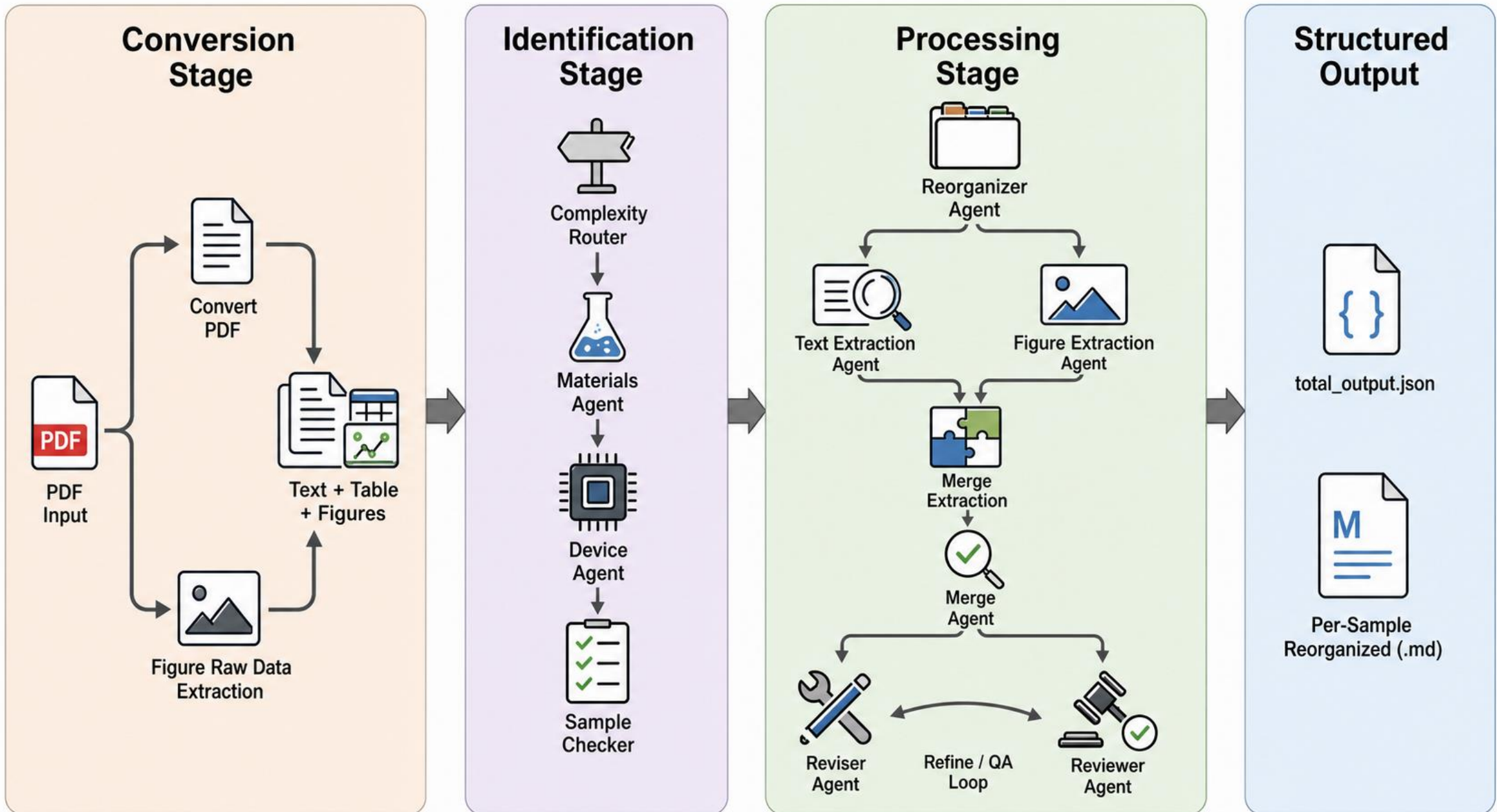
- ✓ Evidence is fragmented across multimodal sources
- ✓ Experimental sweeps create ambiguous sample boundaries
- ✓ Properties must be linked to the correct material or device
- ✓ Figures often contain values absent from text

→ Contextra reorganizes papers into sample-level knowledge



Methods

Workflow



Sample Identification

Study-Structure Parsing

- ✓ Identify candidate materials and devices from paper evidence
- ✓ Resolve sample boundaries using sweep-aware identification rules
- ✓ Lock canonical sample IDs before property extraction
- ✓ Route extracted properties to the matched sample and measurement context

Benchmark Setup

- ✓ Compared with human-curated lists from 14 randomly sampled MSE papers

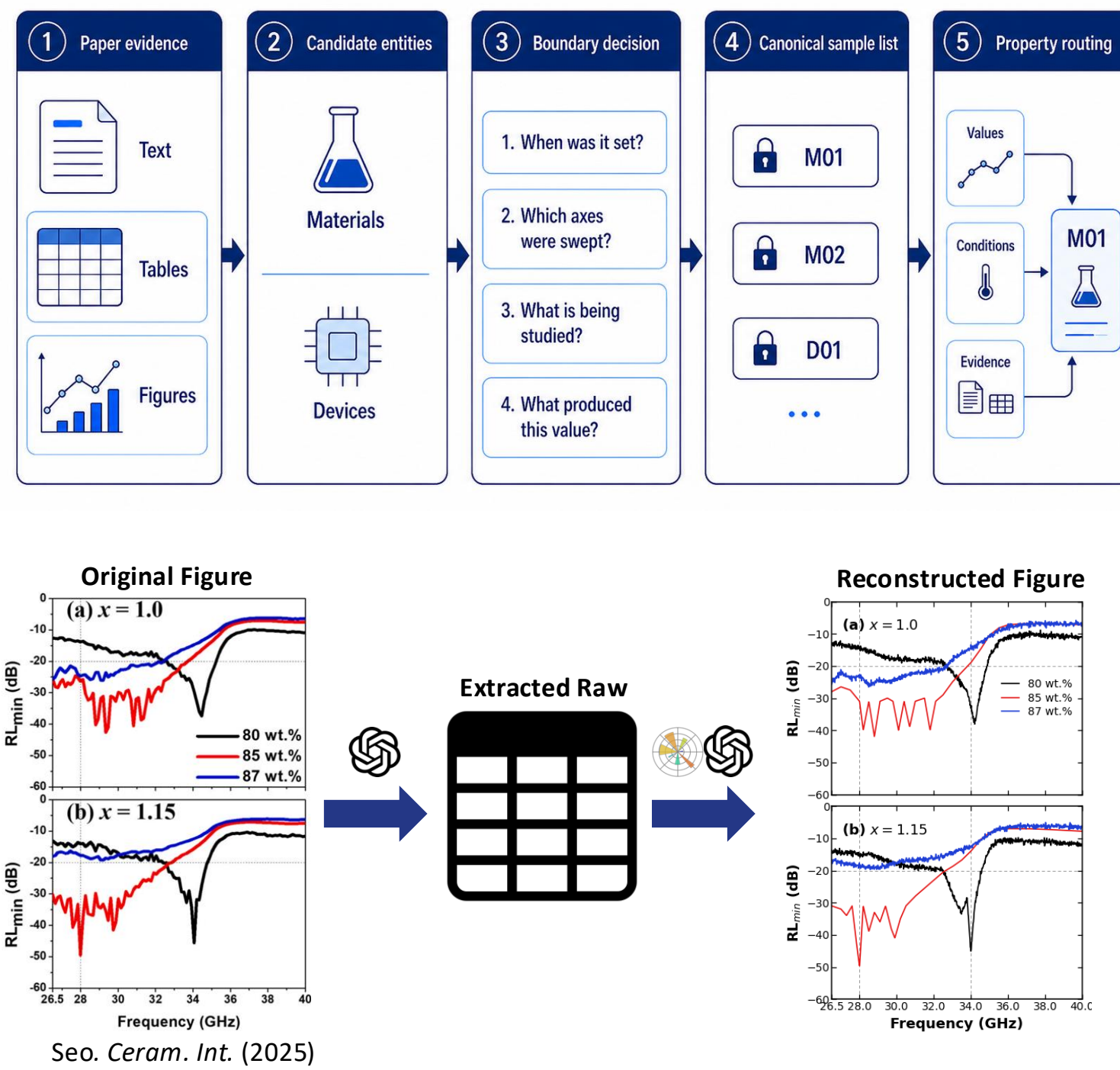
Figure-to-Data Extraction

Figure Parsing

- ✓ Split figures into panels and identify plot regions
- ✓ Extract axes, units, legends, labels, and data points
- ✓ Retrieve numerical data from plots

Benchmark Setup

- ✓ Compared with reconstructed test data from 20 figures



Results

Sample Identification

Prompt Design for Sample Branching

Observed Failure	Prompt Refinement	Example
Measurement over-split	When was it set? Primary subject → sample Measurement vehicle → condition	Films annealed at 300 / 400 / 500 °C → 3 distinct samples One film measured at 25 / 50 / 100 °C → 1 sample, 3 measurement conditions
Experimental variants over-merged	Which axes were swept? Enumerate experimental axis first Measured Combinations → Samples	Composition: A / B / C Annealing: 300 / 500 °C → 6 sample candidates
Measurement vehicle over-added	What is being studied? Primary subject → sample Vehicle / fixture → condition	Cathode material tested in coin cell → cathode material = sample → coin cell = measurement condition
Material-device boundary mixed	What produced this value? Standalone specimen → material Operating assembly → device	Film property before integration → material sample Efficiency / impedance from operated cell → device sample

Performance

- ✓ Recall = matched extracted samples / human samples
- ✓ Precision = matched extracted samples / extracted samples

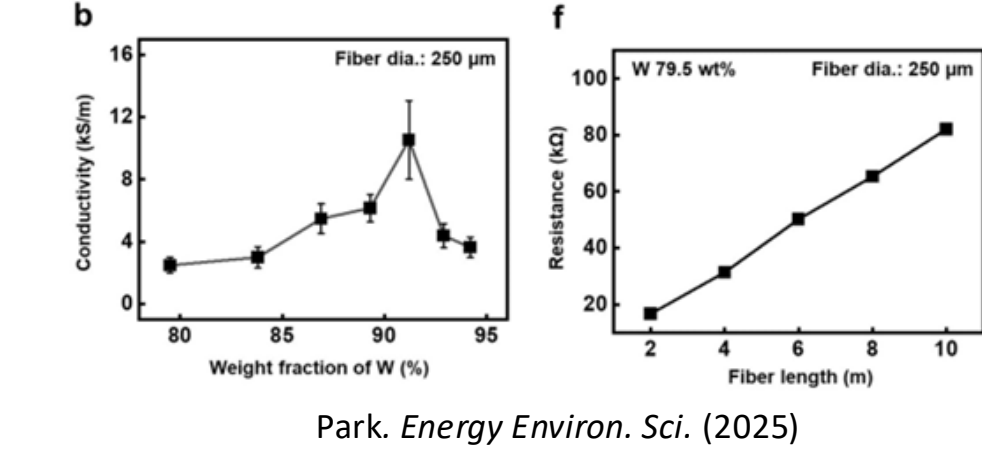
* 3 runs until sample list completion

DOI	Research field	# of sample		Text only		Total (Text + figure)	
		Text only	Total (Text + figure)	Recall avg ± std	Recall avg ± std	Precision avg ± std	Precision avg ± std
10.1007/s13391-024-00508-w	Carbon thin film	4	6	1.00 ± 0.00	0.78 ± 0.16	1.00 ± 0.00	1.00 ± 0.00
10.1016/j.ceramint.2025.10.129	Hexaferrite	17	17	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
10.1016/j.jmat.2025.101101	Ferroelectric device	15	41	0.89 ± 0.11	0.88 ± 0.16	0.96 ± 0.03	0.96 ± 0.03
10.1016/j.polymer.2025.128992	PAN fiber	8	14	1.00 ± 0.00	1.00 ± 0.00	0.92 ± 0.06	0.92 ± 0.06
10.1016/j.snb.2025.139005	Heat sensor	6	9	1.00 ± 0.00	1.00 ± 0.00	0.68 ± 0.16	0.68 ± 0.16
10.1016/j.snb.2026.139699	2D perovskite	5	25	1.00 ± 0.00	0.65 ± 0.25	1.00 ± 0.00	1.00 ± 0.00
10.1021/jacs.4c10143	Chiral material	5	14	0.87 ± 0.09	0.88 ± 0.03	0.93 ± 0.00	0.93 ± 0.00
10.1038/s41467-022-35287-1	Li battery	5	5	1.00 ± 0.00	1.00 ± 0.00	0.66 ± 0.04	0.66 ± 0.04
10.1038/s41528-025-00448-x	Bio electronics	7	15	0.71 ± 0.00	0.47 ± 0.00	0.64 ± 0.00	0.64 ± 0.00
10.1039/d2ee01813f	fuel cell	10	10	1.00 ± 0.00	1.00 ± 0.00	0.89 ± 0.08	0.89 ± 0.08
10.1039/d5ee02650d	solar cell	6	7	0.89 ± 0.16	0.86 ± 0.12	0.82 ± 0.05	0.82 ± 0.05
10.1088/1361-665X/ab5e53	Memory polymer	4	4	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
10.1088/2631-6331/adb2d8	Fiber compound	4	4	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
10.1126/science.aad1818	PVSK diode	13	14	1.00 ± 0.00	0.95 ± 0.03	0.93 ± 0.00	0.93 ± 0.00
Mean value		7.8	13.2	0.95	0.89	0.89	0.89

- ✓ Sweep-aware rules reorganize ambiguous experimental variants into stable sample-level identities

Figure Property Extraction

Unannotated Data vs. Text

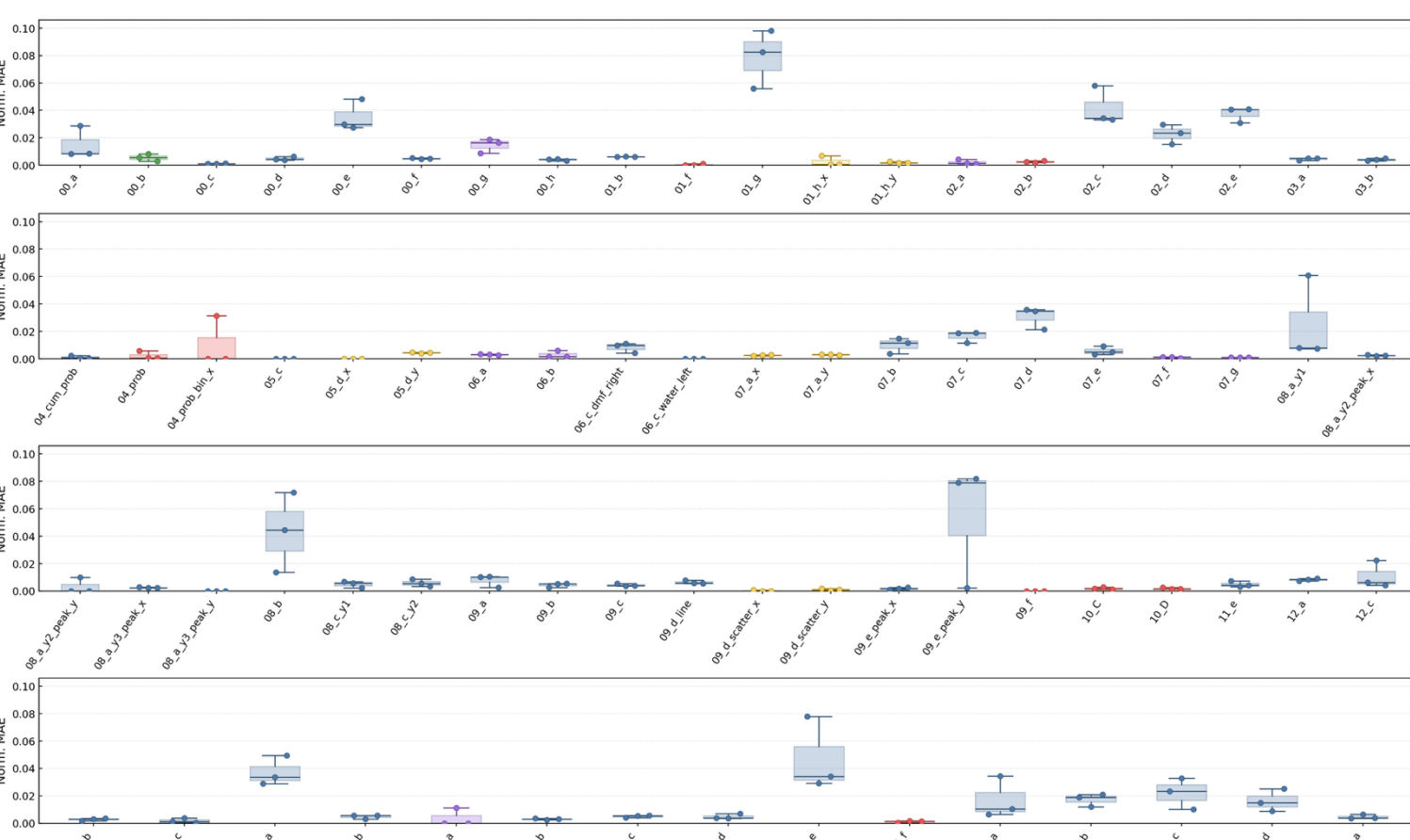
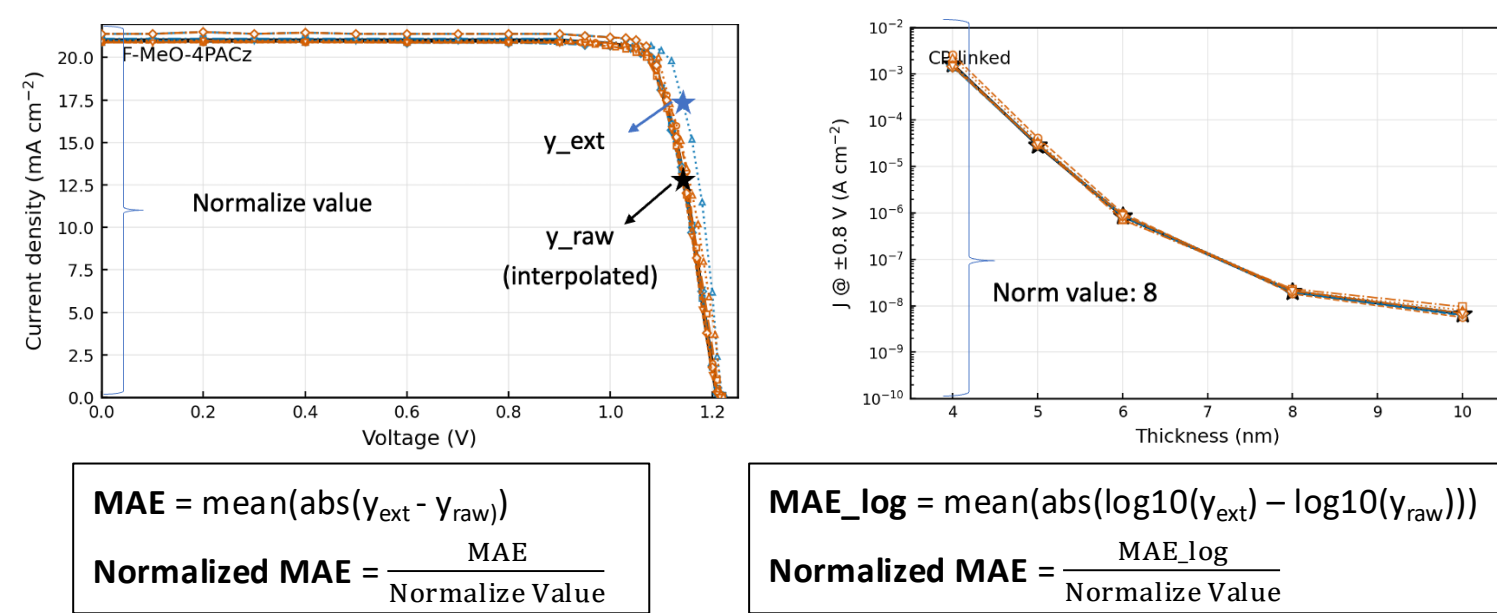


conductivity
• text_span [S/m]
M02 [79.5 wt%]
→ 2200

• figure_unannotated [kS/m]
M02
→ 2.4 (from (b), raw)
→ 2.48 (from (f), calculated)

- ✓ Figure extraction derives text-consistent and calculated properties from plot data.

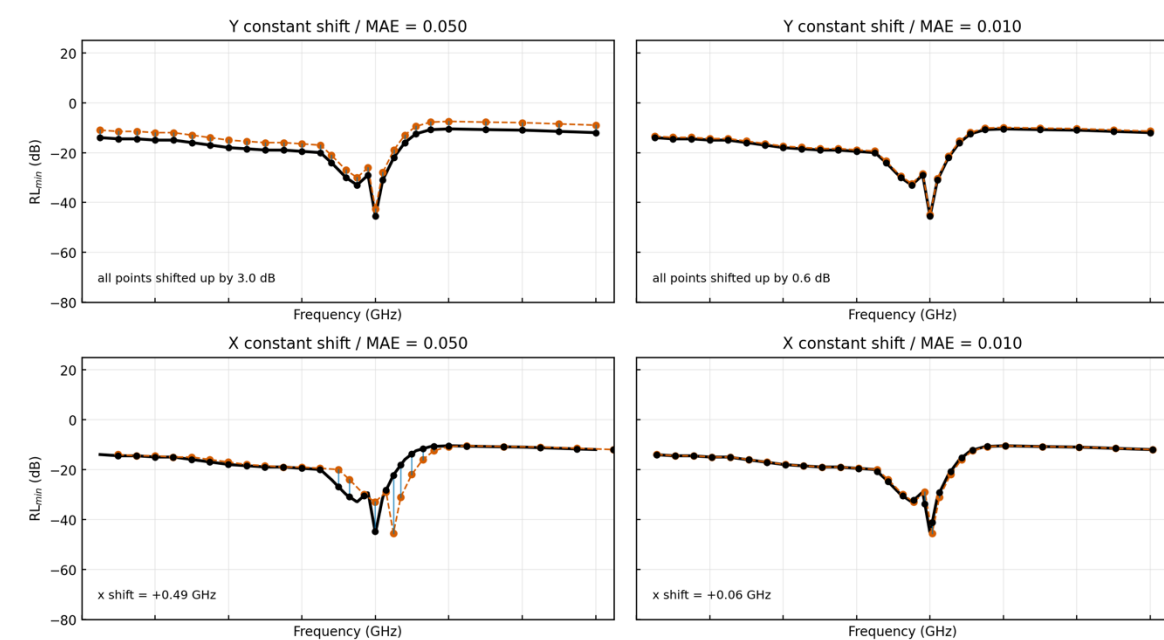
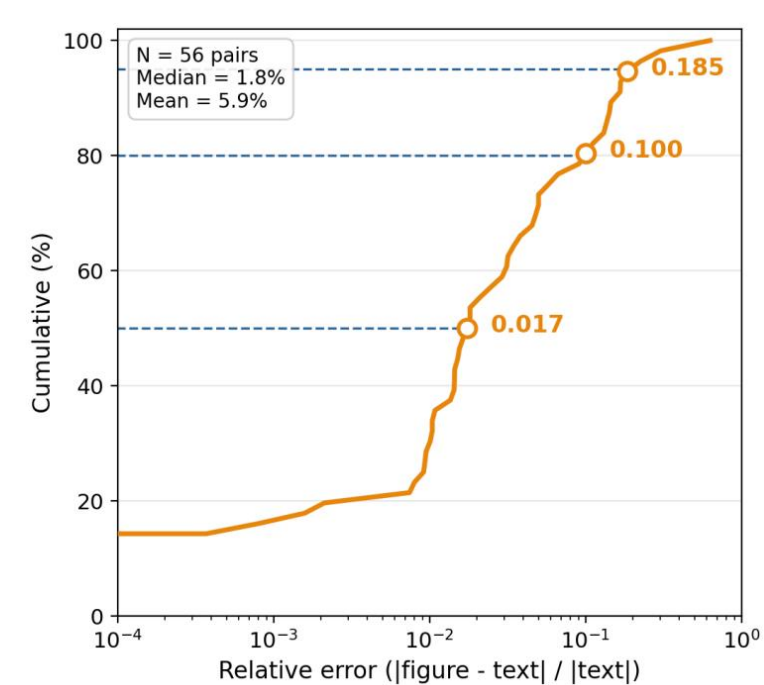
Normalized MAE



Novel Value Rate(NVR)

	Total	Novel	NVR
All properties	939	584	62.2%
Excluding XRD peaks	770	400	52%

- ✓ More than half of the extracted properties were figure-only
- ✓ Figure extraction reveals quantitative knowledge beyond text.



High-fidelity & Robust extraction on plots

- ✓ Normalized MAE ≤ 0.05 corresponds to visually minor deviations, indicating high-fidelity extraction
- ✓ Most test cases show small extraction error.
- ✓ Larger errors are limited to sharp peaks or steep transitions.

Conclusion

- Sweep-aware identification enables reliable sample-level reorganization of materials papers.
- Figure-to-data extraction converts plots into quantitative property evidence, capturing values that text-only analysis would miss.
- The framework transforms scattered literature into structured, evidence-linked materials knowledge.